



AI driven discovery of bio ecological mediation in cascading heatwave risks

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ABSTRACT

Compound heatwaves increasingly trigger cascading failures that propagate through interconnected physical and human systems, yet the fragmentation of disciplinary knowledge hinders comprehensive mapping of these systemic risk topologies. This study introduces the Heatwave Discovery Agent (HeDA), an autonomous scientific synthesis framework designed to bridge disciplinary gaps by constructing a knowledge graph from 8365 academic publications. The system structures 34,933 entities and 42,890 directed relationships from a temporally partitioned corpus (≤ 2024 , $n = 5,739$). On a temporally held-out multi-hop reasoning benchmark (2025–2026, $n = 3,892$), knowledge-graph augmentation yields a consistent accuracy improvement of 8.1% to 9.8% over four standalone foundation-model baselines (McNemar exact test, $p < 10^{-19}$). Topological analysis identifies a bio-ecological mediation effect in which biological systems function as the primary structural mediators of thermal stress, transforming physical meteorological hazards into systemic socioeconomic losses. We further identify latent functional couplings between theoretically distinct sectors, such as the heat-induced synchronization of power grid failures and emergency medical capacity saturation. These findings provide a structural basis for shifting adaptation strategies from static sectoral defense to dynamic cross-system resilience.

1. Introduction

Anthropogenic forcing has substantially reshaped the global risk landscape, where the aggregation of rainfall extremes and the intensification of oceanic heat potential increasingly converge with thermal anomalies to create compound hazards (Konda et al., 2024; Vissa et al., 2013a,b). Within these evolving thermodynamic regimes, heatwaves have emerged as one of the most lethal and socioeconomically devastating hazards facing humanity. The 2003 European heatwave caused over 70,000 deaths and extensive economic losses (Robine et al., 2008; García-Herrera et al., 2010; Beniston, 2004; Black et al., 2004), while the 2021 Pacific Northwest heatwave broke historical temperature records and triggered cascading failures across infrastructure systems such as power grids and emergency services (White et al., 2023; Saki et al., 2025). Unlike discrete natural disasters, heatwaves create multidomain cascading effects that propagate through interconnected physical, social, and economic systems in ways that traditional risk assessment approaches often struggle to capture (Ebi, 2025). As these compound events become more frequent, the inability to foresee how environmental stress transmits to human vulnerability represents a significant gap in current capacity for climate adaptation.

This challenge is exacerbated by the fragmentation of scientific knowledge, where important risk insights represent long-tail knowledge buried within thousands of disconnected studies spanning climatology, public health, and economics. These specific cascading pathways appear infrequently in general training corpora, causing standard large language models to often struggle with retrieval and reasoning tasks. Without domain-specific fine-tuning or external knowledge grounding, even state-of-the-art foundation models often hallucinate or overlook these rare connections (Reichstein et al., 2025). While artificial intelligence offers promising tools for synthesis, existing reactive systems lack the ability to capture and reason over this sparse long-tail information to identify high-impact vulnerabilities that researchers have not yet hypothesized (Ramachandra, 2025; Al Khouardj, 2025).

We present the Heatwave Discovery Agent (HeDA), an intelligent multi-agent system designed to bridge this gap through automated scientific discovery and multi-layer risk propagation analysis. By processing 8365 academic papers, HeDA constructs a knowledge graph containing 34,933 nodes and 42,890 relationships from a temporally partitioned corpus (≤ 2024 , $n = 5739$), which serves as a structured repository (Peng et al., 2023) for capturing long-tail scientific evidence. Evaluated as a model-agnostic framework on a temporally

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held-out multi-hop benchmark ($n = 3892$), KG-augmented inference yields statistically significant accuracy gains of +8.1% to +9.8% across four foundation models (Qwen-Plus, Qwen3-Max, GPT-5.4, and Claude-Sonnet-4.5; all McNemar $p < 10^{-19}$), with the strongest effect at 2-hop depth (from +13.7% to +17.9%), demonstrating that structured knowledge graphs complement parametric reasoning without model-specific fine-tuning.

Beyond technical performance, our network analysis identifies a bio-ecological mediation effect in which biological systems function as the primary structural mediators of thermal stress, transforming physical hazards into systemic economic losses. HeDA further uncovers latent functional couplings between theoretically distinct infrastructure sectors and identifies high-impact grey rhino risk chains that represent structurally plausible but underexplored transmission vectors. These findings illustrate the potential of artificial intelligence to decipher the anatomy of cascading climate risks and provide a structural basis for shifting adaptation strategies from static defense to dynamic system resilience.

2. Related work

2.1. Compound weather extremes and systemic risk propagation

The characterization of climate hazards has shifted from a univariate perspective to a focus on compound weather and climate events where the combination of multiple drivers contributes to societal or environmental risk (Zscheischler et al., 2018, 2020). Unlike isolated meteorological anomalies, these concurrent or sequential stressors, including aerosol-enhanced precipitation events and shifting pollution hotspots (Choudhury et al., 2020; Tyagi et al., 2021), generate non-linear cascading effects that propagate through interconnected physical and human systems (Raymond et al., 2020; Niggli et al., 2022). Historical analyses of the 2003 European heatwave and the 2021 Pacific Northwest event indicate that the interaction between atmospheric blocking patterns and surface thermodynamic exchanges involving sensible and latent heat fluxes (Bandaray et al., 2026; Tyagi et al., 2013) creates conditions where thermal stress acts as a systemic multiplier rather than a discrete hazard (Robine et al., 2008; White et al., 2023). Current research underscores that these cascading failures often transcend geographical and sectoral boundaries thereby challenging traditional risk assessment frameworks that treat physical hazards and socioeconomic impacts as linearly related variables (Simpson et al., 2021; Westra and Zscheischler, 2023).

2.2. Knowledge fragmentation and the limits of disciplinary modeling

Despite the growing recognition of systemic risks, the capability to model cross-sectoral cascading pathways remains constrained by the fragmentation of disciplinary knowledge. Integrated Assessment Models and standard coupled physical-biogeochemical frameworks frequently struggle to capture the complex adaptive nature of terrestrial ecosystems and their feedback to the climate system (Ackerman et al., 2009; Frank et al., 2015). Studies indicate that single-sector impact models systematically misrepresent the magnitude of risks because they omit the interdependencies between water resources, agriculture, and energy systems (Harrison et al., 2016; Monier et al., 2018). The knowledge required for mapping these complex topologies is distributed across disparate fields ranging from atmospheric physics to epidemiology and public health, which creates significant blind spots in identifying the biological and ecological mediators that govern risk transmission (Talukder et al., 2024; Naylor et al., 2020). This disconnect highlights the need for synthesis methodologies capable of identifying latent structural dependencies that are currently obscured by the sheer volume and disciplinary segregation of scientific evidence.

2.3. AI-driven synthesis and knowledge discovery in earth science

Addressing the challenge of information fragmentation necessitates the deployment of computational tools capable of synthesizing vast unstructured scientific corpora into computable formats. The emergence of artificial intelligence for scientific discovery offers a promising approach to reconstructing these complex risk topologies by moving beyond simple information retrieval to high-dimensional hypothesis generation (Wei et al., 2025; Reddy and Shojaaee, 2025). Recent advancements in large language models and structured information extraction facilitate the automated construction of high-fidelity knowledge graphs that can resolve terminological heterogeneity across disciplines and identify mechanistic linkages hidden in long-tail literature (Dagdelen et al., 2024; Hartung, 2025). By integrating neuro-symbolic reasoning with domain-specific constraints, these autonomous systems provide a capability to bridge the gap between physical climate dynamics and socioeconomic vulnerability, effectively functioning as a macroscope for detecting emergent systemic risks that human researchers may overlook (Zhu et al., 2024; Y. Li et al., 2025).

3. Methods

3.1. Multi-agent system architecture

To overcome the inherent limitations of traditional disciplinary modeling in capturing cross-sectoral cascading risks, we developed HeDA as an autonomous multi-agent framework for analyzing Earth system interactions. Unlike standard information retrieval tools, which treat scientific evidence as static text strings, HeDA functions as a structured synthesis framework that reconstructs the non-linear topology of heatwave risks by integrating fragmented mechanisms from climatology, ecology, and economics. The system architecture orchestrates a hierarchical multi-agent workflow where a central coordination unit manages the dynamic scheduling of tasks across extraction construction and reasoning modules as illustrated in Fig. 1. This architecture explicitly addresses the challenge of scale mismatch in climate risk assessment by transforming unstructured qualitative descriptions from the abstracts of 8365 documents into a computable directed graph containing 34,933 nodes and 42,890 edges. By structuring evidence through this standardized protocol HeDA enables the quantitative analysis of how physical atmospheric anomalies propagate through biological mediators to generate systemic socioeconomic losses.

The synthesis process is organized into a seven-stage protocol designed to ensure the physical plausibility and semantic consistency of the constructed knowledge graph. The workflow begins with corpus acquisition and filtering, where relevance verification algorithms screen the abstracts to remove noise and establish a reliable baseline. The subsequent ontological extraction stage uses large language models to identify relational triplets while preserving the conditional boundaries of scientific claims. To resolve the terminological heterogeneity inherent in cross-disciplinary research, the system executes semantic disambiguation using sentence transformers to cluster synonymous entities, such as categorizing thermal stress and heat exposure into unified nodes. Next, attribute enrichment classifies edges into physical, social, or economic layers, followed by topological construction within a graph database environment. The sixth stage involves vector embedding to map discrete entities into a continuous latent space facilitating semantic retrieval. Finally, the reasoning and verification stage applies graph traversal algorithms to identify latent risk pathways that span multiple systemic layers and assesses their structural coherence against established scientific principles.

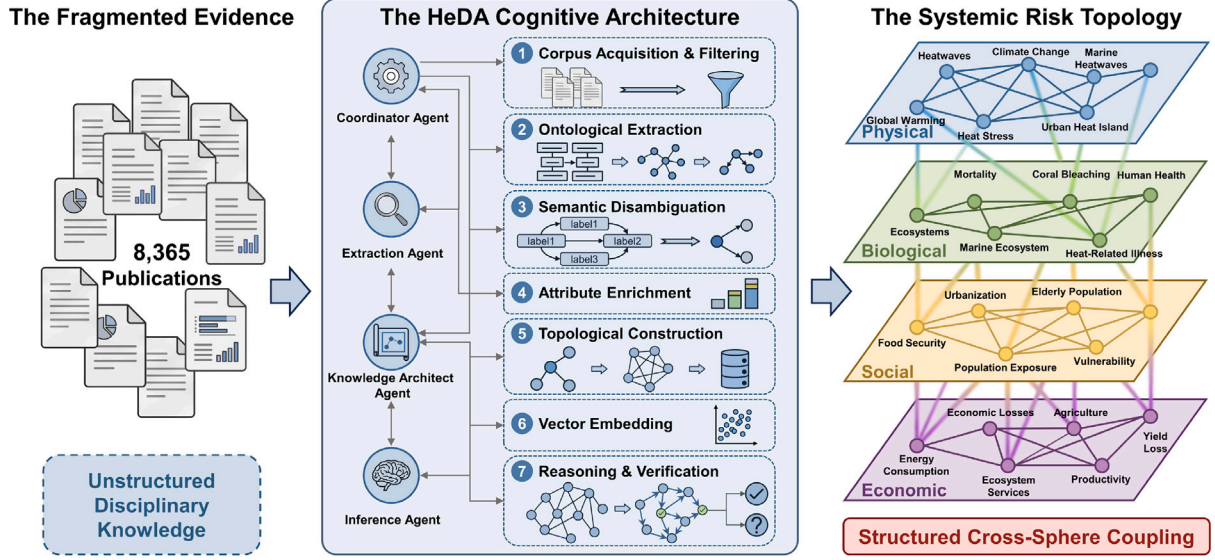


Fig. 1. Architectural workflow of the Heatwave Discovery Agent (HeDA). The system transforms 8365 fragmented publications into a structured risk topology through a seven-stage neuro-symbolic protocol: (1) Corpus Acquisition & Filtering, (2) Ontological Extraction, (3) Semantic Disambiguation, (4) Attribute Enrichment, (5) Topological Construction, (6) Vector Embedding, and (7) Reasoning & Verification. Four coordinated agents (Coordinator, Extraction, Knowledge Architect, Inference) manage the pipeline, producing a four-layer systemic risk topology (Physical, Biological, Social, Economic) that captures structured cross-sphere coupling from unstructured disciplinary knowledge.

3.2. Multi-layer risk propagation analysis

We establish a mathematical formalism for identifying latent systemic vulnerabilities by treating the risk landscape as a multi-layered topology where failures propagate across distinct system layers. This framework shifts the analytical focus from static node attributes to the dynamic nature of interaction edges effectively capturing the flow of cascading risks through the definition of a mapping function L . This function categorizes relationships into four fundamental systemic layers comprising physical climatic processes, biological ecosystem responses, social vulnerability, and economic dependencies.

$$L : E \rightarrow \{\text{Physical, Biological, Social, Economic, CrossLayer}\}$$

$$\text{where } L(e) = \begin{cases} \text{Physical} & \text{if } e \in \{\text{climatic anomalies, hydrological cycles}\} \\ \text{Biological} & \text{if } e \in \{\text{ecosystem functions, physiological health}\} \\ \text{Social} & \text{if } e \in \{\text{public health, labor capacity, migration}\} \\ \text{Economic} & \text{if } e \in \{\text{infrastructure, market dynamics, GDP}\} \end{cases}$$

(1)

All four layers are retained as independent categories throughout the primary analysis. We quantify the structural role of biological systems as a mediation bridge via the Bio-ecological Mediation Ratio:

$$\text{BMR} = \frac{f_{P \rightarrow B} + f_{B \rightarrow E}}{f_{P \rightarrow E}} \quad (2)$$

where $f_{X \rightarrow Y}$ denotes the number of directed edges from layer X to layer Y . A BMR substantially greater than one indicates that indirect transmission through the biological layer dominates over the direct physical-to-economic shortcut. The sensitivity of this metric to alternative layer groupings (e.g., collapsing Social and Economic into a single socioeconomic domain) is evaluated in Section 4.5. To systematically uncover high impact pathways that remain obscured in fragmented literature we define a structural novelty metric for any given propagation chain P . This metric operationalizes the concept of the grey rhino risk by synthesizing three theoretical components into a unified score.

$$\text{NoveltyScore}(P) = \alpha \cdot \text{LF}(P) + \beta \cdot \text{CLC}(P) + \gamma \cdot \text{IP}(P) \quad (3)$$

The Literature Frequency component $\text{LF}(P)$ acts as an inverse popularity filter to identify rare co-occurrences that represent information-theoretic novelty relative to the training corpus, defined by $1 - \frac{f(P)}{F_{\max}}$. The Cross-Layer Connectivity component $\text{CLC}(P)$ quantifies the topological complexity of the risk chain by measuring the density of domain transitions using an indicator function $\mathbb{I}$ that counts transitions across system boundaries.

$$\text{CLC}(P) = \frac{\sum_{i=1}^{n-1} \mathbb{I}(L(r_i) \neq L(r_{i+1}))}{n-1} \quad (4)$$

The Impact Potential $\text{IP}(P)$ integrates the structural centrality of nodes with the confidence levels of extracted relations to ensure that discovered pathways represent structurally cohesive transmission vectors rather than isolated literature co-occurrences. We calibrate the weighting parameters α , β , and γ to prioritize pathways that exhibit both high structural novelty and significant propagation potential.

3.3. Risk discovery algorithm and implementation

Compound heatwaves exert non-linear cascading effects that transcend the resolution boundaries of traditional coupled physical-biogeoclimatic models, where the propagation of vulnerability involves complex feedback loops between atmospheric anomalies and human systems. The interactions between meteorological stressors and socioeconomic losses are rarely direct but are modulated through intermediate ecological buffers that remain computationally opaque in standard sectoral assessments. To address this limitation, we developed a multi-hop risk discovery algorithm that treats the knowledge graph as a high-dimensional state space where cascading failures are modeled as directed propagation paths constrained by specific depth and layer-transition parameters. The inference engine implements a parallelized graph traversal strategy that filters out shallow associative correlations while prioritizing deep structural dependencies that traverse multiple system boundaries from physical hazards through biological systems to economic outcomes. By integrating a structural novelty metric with structural consistency verification, the system isolates high-impact low-frequency transmission vectors located in the long tail of the probability distribution, which characterizes the anatomy of grey rhino risks. This

computational framework synthesizes the processed evidence nodes derived from the academic corpus into a unified risk topology that allows for the quantitative identification of the bio-ecological mediation layer as a key bottleneck in systemic failure propagation.

Algorithm 1 Multi-Layer Risk Propagation Discovery

Require: Knowledge graph $G = (V, E)$, layer mapping L , $\min_hops$
 $d_{\min} = 2$, $\max_hops$ $d_{\max} = 5$

Ensure: Set of high-novelty risk pathways $\mathcal{P}$

- 1: Initialize $\mathcal{P} \leftarrow \emptyset$, $visited \leftarrow \emptyset$
- 2: Collect all layer-classified nodes $S \leftarrow \{v \in V \mid L(v) \in \{P, B, S, E\}\}$
- 3: Randomly shuffle S and partition into W parallel workers
- 4: **for** each source node $s \in S$ in parallel **do**
- 5: $queue.enqueue(\{s\}, 0)$
- 6: **while** $queue \neq \emptyset$ **do**
- 7: $(path, depth) \leftarrow queue.dequeue()$
- 8: **if** $depth < d_{\max}$ **then**
- 9: $current \leftarrow path.last()$
- 10: **for** each neighbor n of $current$ **do**
- 11: $new_path \leftarrow path.append(n)$
- 12: **if** $depth + 1 \geq d_{\min}$ AND new_path spans ≥ 2 distinct layers **then**
- 13: Add new_path to $\mathcal{P}$
- 14: **end if**
- 15: **if** $new_path \notin visited$ **then**
- 16: $visited.add(new_path)$
- 17: $queue.enqueue(new_path, depth + 1)$
- 18: **end if**
- 19: **end for**
- 20: **end if**
- 21: **end while**
- 22: **end for**
- 23: Deduplicate $\mathcal{P}$
- 24: **for** each $path \in \mathcal{P}$ **do**
- 25: $score \leftarrow NoveltyScore(path) \{\alpha \cdot LF + \beta \cdot CLC + \gamma \cdot IP\}$
- 26: **end for**
- 27: **return** Pathways from $\mathcal{P}$ ranked by novelty score

The pathways identified by this algorithm represent structural patterns in the literature topology, not validated causal mechanisms. Cross-layer transitions reflect co-occurrence and directed argumentation within published studies; establishing causality would require independent interventional or observational designs (e.g., structural equation modeling, natural experiments). We therefore interpret high-novelty paths as structurally grounded hypotheses warranting further empirical investigation.

3.4. Dataset construction and technical infrastructure

To ensure comprehensive coverage of the heatwave risk landscape we systematically retrieved 8365 academic records from the Web of Science Core Collection (topic query: “heatwave”, retrieval date: 8 April 2026) spanning the period from 1928 to 2026. The corpus was temporally partitioned into a knowledge-graph construction set (≤ 2024 , $n = 5739$) and a held-out evaluation set (≥ 2025 , $n = 2626$), ensuring zero temporal overlap between training evidence and benchmark questions. Relational triplets were extracted from each abstract using Claude Sonnet 4.5 under a structured ontological prompt that enforces four-layer classification (Physical, Biological, Social, Economic) and requires verbatim evidence sentences; an exponential-backoff retry mechanism (up to three attempts with 2^k -second delays) ensured robustness against API rate limits. Records lacking sufficient abstract content (approximately 3.5%, $n = 30$) returned empty arrays rather than hallucinated triples. Synonymous entities were merged via cosine-distance clustering on text-embedding-v4 vectors (DashScope) at a conservative threshold

of $d < 0.03$ (cosine similarity > 0.97), yielding a directed graph of 34,933 nodes and 42,890 edges. The sensitivity of downstream conclusions to this threshold is evaluated in Section 4.5. All graph operations use NetworkX; semantic retrieval is powered by FAISS vector indexing.

To evaluate the knowledge graph’s utility for multi-hop reasoning without circular self-evaluation, we constructed the Heatwave Risk Reasoning Benchmark (HRRB) exclusively from the ≥ 2025 held-out corpus. A unified cross-paper graph was first assembled from the held-out triples (19,526 nodes, 23,537 edges). Random-walk sampling over this graph produced 1–4 hop paths, each of which was fed to an LLM that reverse-generated a multiple-choice question whose ground-truth answer is anchored to the source triple and its verbatim evidence sentence, ensuring that the correct answer is independently verifiable from a published paper rather than from the knowledge graph itself. Distractors were generated by the same LLM as semantically plausible but factually incorrect alternatives, avoiding the trivially distinguishable node-permutation approach.

Crucially, the benchmark is formalized as a Latent Link Prediction task under strict chronological constraints to ensure it evaluates scientific discovery rather than information retrieval. Let $G_{\text{base}} = (V_{\text{base}}, E_{\text{base}})$ denote the historical knowledge graph constructed from the training corpus ($t \leq 2024$), and $G_{\text{future}} = (V_{\text{future}}, E_{\text{future}})$ represent the ground-truth topology derived exclusively from the held-out literature ($t \geq 2025$). For any target question mapped to an n -hop cascading path $P_n = \{e_1, e_2, \dots, e_n\} \subseteq E_{\text{future}}$, the generation algorithm enforces a Temporal Discovery Constraint $\mathcal{T}(P_n)$:

$$\mathcal{T}(P_n) = \begin{cases} P_n \subseteq E_{\text{base}} & \text{if } n = 1 \\ 0 < \sum_{i=1}^n \mathbb{I}(e_i \in E_{\text{base}}) < n & \text{if } n \geq 2 \end{cases} \quad (5)$$

where $\mathbb{I}$ is the indicator function. This formulation enforces two evaluation properties. For the single-hop case ($n = 1$), questions are restricted to facts already established in the historical baseline (E_{base}), testing parametric memory where knowledge-graph augmentation is expected to yield minimal discriminative gain. The 1-hop pool is consequently smaller ($n = 892$) because it draws only from the intersection $E_{\text{base}} \cap E_{\text{future}}$. For multi-hop questions ($n \geq 2$), pathways must contain a structural mixture of known historical edges and at least one novel edge ($e_i \notin E_{\text{base}}$) first formally reported in 2025 or later. Correctly answering such a question requires the model to use historical topological skeletons to logically deduce emergent cross-disciplinary cascading mechanisms formalized years later, thereby performing latent link prediction across disciplinary boundaries and eliminating any risk of evaluation circularity.

Under the knowledge-augmented (+KG) condition, each question entity is linked to the ≤ 2024 KG via FAISS nearest-neighbor search on text-embedding-v4 vectors (cosine similarity threshold 0.82). Two breadth-first searches (cutoff = 4 hops) are then launched from the start and end entities on an undirected projection of the KG. For every candidate answer option, the retriever reports the shortest-path connectivity and relationship labels to both endpoints (classified as DIRECT, 2–3 steps, or unreachable) as well as local neighborhood triples (up to five successors and three predecessors of each endpoint). This structured evidence block is prepended to the LLM prompt, which instructs the model to prioritize options with strong graph connectivity while falling back to domain knowledge when the KG is silent.

The final benchmark comprises $N = 3892$ questions stratified by hop depth: 1-hop ($n = 892$, within-layer negative control), 2-hop ($n = 1000$), 3-hop ($n = 1000$), and 4-hop ($n = 1000$), with answer positions uniformly balanced across A/B/C/D via round-robin assignment. During evaluation, a strict format constraint (“Respond with ONLY a single letter”) was appended to the prompt, achieving 100% format compliance across all model responses and eliminating parsing artifacts.

Table 1

Multi-Hop LBD Benchmark results ($N = 3,892$). Four frontier LLMs evaluated under standalone (No-KG) and knowledge-augmented (+KG) conditions. Δ rows report the absolute accuracy gain; p values are from McNemar's exact test (two-sided).

Model	Mode	1-hop (%)	2-hop (%)	3-hop (%)	4-hop (%)	Accuracy (%)	F1-macro	p (McNemar)
Qwen-Plus	No-KG	42.38	58.00	55.00	48.40	51.18	0.512	–
	+KG	45.63	71.70	58.10	60.20	59.28	0.593	–
	Δ	+3.25	+13.70	+3.10	+11.80	+8.09	+0.081	3.09×10^{-20}
Qwen3-Max	No-KG	43.39	58.30	56.50	48.70	51.95	0.519	–
	+KG	47.09	74.60	61.10	60.40	61.18	0.612	–
	Δ	+3.70	+16.30	+4.60	+11.70	+9.22	+0.092	1.86×10^{-27}
GPT-5.4	No-KG	43.05	56.50	51.30	44.70	49.05	0.490	–
	+KG	46.41	74.40	56.20	56.80	58.79	0.588	–
	Δ	+3.36	+17.90	+4.90	+12.10	+9.74	+0.097	2.93×10^{-25}
Claude-Sonnet-4.5	No-KG	41.03	58.50	52.20	47.60	50.08	0.498	–
	+KG	47.98	75.50	59.60	55.20	59.89	0.595	–
	Δ	+6.95	+17.00	+7.40	+7.60	+9.82	+0.097	8.72×10^{-28}

4. Results

4.1. Topological reconstruction of compound risk landscapes

We evaluate the utility of the knowledge graph for multi-hop scientific reasoning using a temporally held-out benchmark. The Multi-Hop Literature-Based Discovery (LBD) Benchmark comprises $N = 3892$ multiple-choice questions stratified by inference depth: 1-hop ($n = 892$, negative control), 2-hop ($n = 1000$), 3-hop ($n = 1000$), and 4-hop ($n = 1000$). All questions were generated from the ≥ 2025 held-out corpus, ensuring zero temporal overlap with the knowledge-graph construction set (≤ 2024). Four frontier LLMs were tested under two conditions, standalone (No-KG) and knowledge-augmented (+KG), with statistical significance assessed via McNemar's exact test (two-sided). As shown in Table 1, the 1-hop factual questions, designed as a negative control where the queried relationship already exists in the ≤ 2024 KG (Eq. (5), $n = 1$), show marginal albeit significant improvements ($\Delta = +3.25\%$ to $+6.95\%$). This is expected, as foundation models already possess strong parametric memory for established single-hop facts and the retriever provides no discriminative advantage over baseline knowledge. The true inferential power of HeDA emerges in multi-hop scenarios, where each question contains at least one novel edge absent from the historical KG (Eq. (5), $n \geq 2$). At 2-hop depth, knowledge-graph augmentation yields its strongest effect (Δ up to $+17.9\%$, all $p < 10^{-15}$), indicating that the retriever successfully supplies relational chains from the historical topology that bridge to novel relationships first formalized in 2025–2026. At 4-hop depth gains remain robust ($\Delta = +7.6\%$ to $+12.1\%$, all $p < 10^{-4}$), confirming that HeDA is not merely retrieving memorized text but using pre-2024 structural topologies to reconstruct cross-disciplinary risk mechanisms formalized years later. At 3-hop depth, where intermediate links are often sparse in the literature, three of four models still show significant improvement; the exception is Qwen-Plus ($p = 0.093$), whose weaker parametric capacity limits its ability to arbitrate between noisy retrieved evidence and internal knowledge. This topological foundation establishes the basis for identifying latent structural vulnerabilities and bio-ecological bottlenecks analyzed in the following sections.

4.2. Bio-ecological amplification mechanisms

The non-linear nature of compound heatwaves requires a departure from linear impact models to a systemic perspective whereby physical hazards propagate through intermediate layers before manifesting as economic deficits. Topological analysis of the constructed knowledge graph identifies a marked structural asymmetry in risk transmission that challenges conventional direct impact assessments by highlighting the role of biological systems as primary structural mediators. Quantitative flux density analysis in Fig. 2 yields a Bio-ecological Mediation Ratio $BMR = 8.147$ (bootstrap 95% CI [7.457, 8.915]), indicating that

indirect transmission through the biological layer is approximately eight times more prevalent than the direct physical-to-economic shortcut. This structural asymmetry indicates that, within the synthesized literature topology, the magnitude of thermal energy is rarely represented as directly translating into economic destruction, but is instead predominantly mediated through biological substrates before manifesting as systemic economic loss. The visualization of these cascading flows in Fig. 3 further elucidates this mechanism by identifying an amplification zone where biological systems, specifically agricultural yield and human physiological health, function as mediating channels of environmental stress. This bottleneck architecture implies that biological systems absorb the initial shock of meteorological extremes and transduce discrete physical parameters into tangible socioeconomic deficits through processes such as crop failure and labor capacity reduction. HeDA identified a latent physiological pathway linking thermal oxidative stress to chronic kidney dysfunction, consistent with the pathogenesis of CKDu observed in agricultural regions (Ben Khadda et al., 2024; Sorensen and Garcia-Trabanino, 2019). Recent projections corroborate this computationally derived finding by estimating that heat-stress-induced labor productivity losses could reduce global agricultural output by 18% by the end of the century (Sheng et al., 2025; Casey et al., 2024). Consequently, the derived network suggests that the stability of the economic layer depends less on the absolute intensity of the heatwave than on the resilience thresholds of intermediate ecological mediators, implying that current infrastructure-centric risk assessments may structurally underestimate systemic vulnerability by neglecting these saturation points.

4.3. Structural synchronization of cross-sectoral vulnerabilities

The systemic risk landscape under extreme thermal regimes exhibits a notable topological restructuring whereby theoretically distinct physical and socioeconomic sectors coalesce into integrated vulnerability complexes. As shown in Fig. 4a, risk entities do not distribute randomly but self-organize into functional clusters that do not align with standard administrative boundaries. This topological synchronization indicates that independent infrastructure systems, such as electrical power grids and emergency medical services, effectively cluster into a shared vulnerability community during heatwaves, as the thermodynamic failure of cooling mechanisms is densely coupled with clinical capacity saturation. A closer examination of the bridge nodes in Fig. 4b provides a deeper explanatory mechanism for this coupling by quantifying the brokerage power of specific hazard entities. Notably, the analysis identifies marine heatwaves and compound drought events as the dominant coupling agents with the highest betweenness centrality scores, rather than purely socioeconomic factors. This structural prominence indicates that compound natural hazards act as the primary topological bridges connecting atmospheric physical anomalies to downstream biological and economic losses. This topological inference is consistent with the 2022

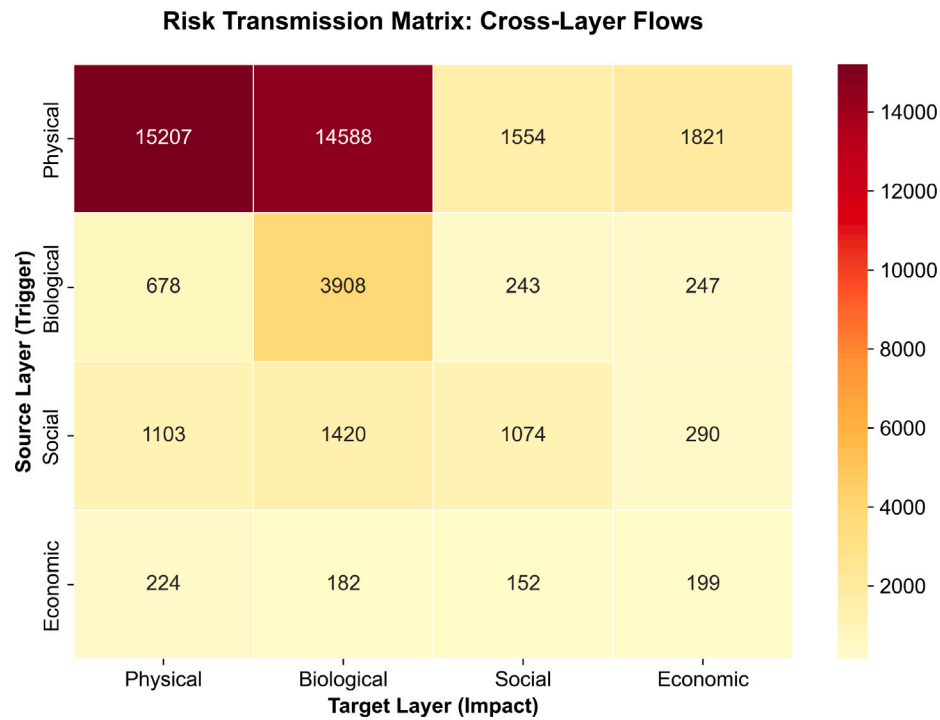


Fig. 2. Risk Transmission Matrix: Cross-Layer Flows. The heatmap quantifies relational density between four systemic layers across 42,890 directed edges, of which 22,502 (52.5%) are cross-layer. The dominant flow from Physical to Biological (14,588 edges) exhibits a 21.5× directional asymmetry over the reverse (678 edges), confirming the unidirectional trigger-to-impact topology. Intra-layer edges account for 47.5% (Physical→Physical: 15,207; Biological→Biological: 3908), reflecting within-domain knowledge consolidation.

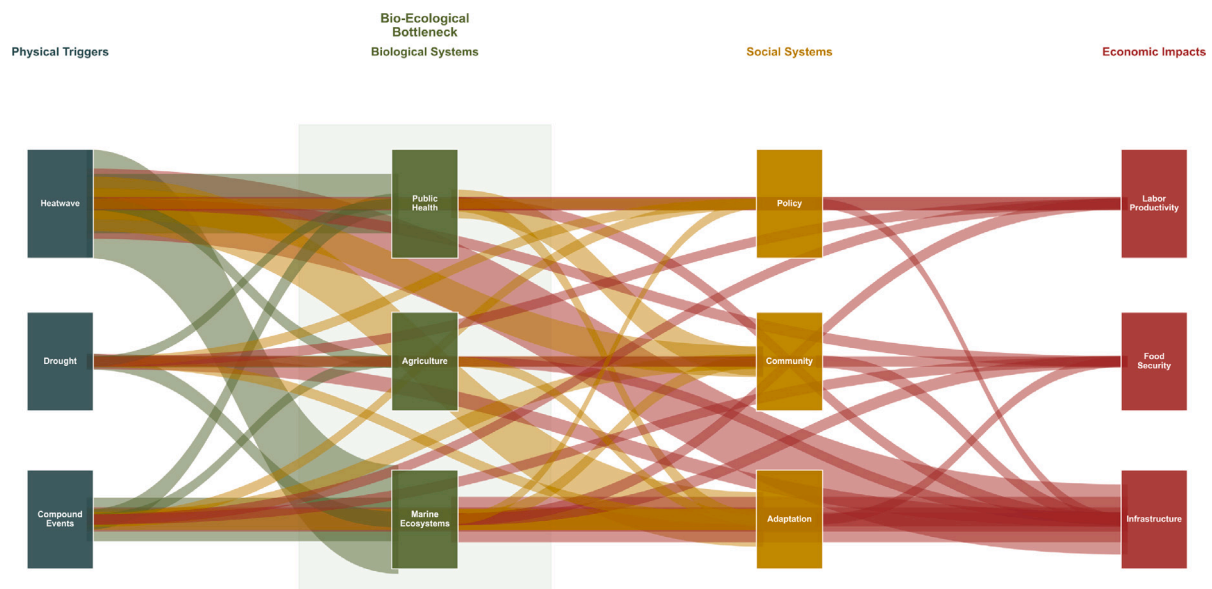


Fig. 3. The Bio-Ecological Bottleneck Architecture. The Sankey diagram decomposes cross-layer risk transmission into sub-category pathways across four independent columns (Physical Triggers → Biological Systems → Social Systems → Economic Impacts). The Physical→Biological weighted flow (3978) exceeds Physical→Economic direct flow (1666) by 2.39×, identifying the biological layer as a structural bottleneck rather than a passive conduit. The strongest sub-category pathway is Heatwave→Marine Ecosystems (1923), followed by Heatwave→Infrastructure (1175) and Heatwave→Public Health (960).

compound drought and heatwave event in the Yangtze River Basin, where the synchronization of hydrological deficits and heat-induced demand surges triggered a 50% reduction in hydropower capacity and consequent industrial shutdowns (J. Li et al., 2025; Liu et al., 2023, 2025). Similar spatial compounding effects involving energy linkages have been projected to intensify across major urban clusters under

future climate scenarios (Fang et al., 2025; Lv et al., 2024). The high ranking of mortality risk and resilience nodes further corroborates the bio-ecological mediation hypothesis, suggesting that the stability of the entire risk network functionally depends on the physiological limits of biological organisms and their capacity to withstand compound stressors. These findings imply that the most dangerous propagation vectors

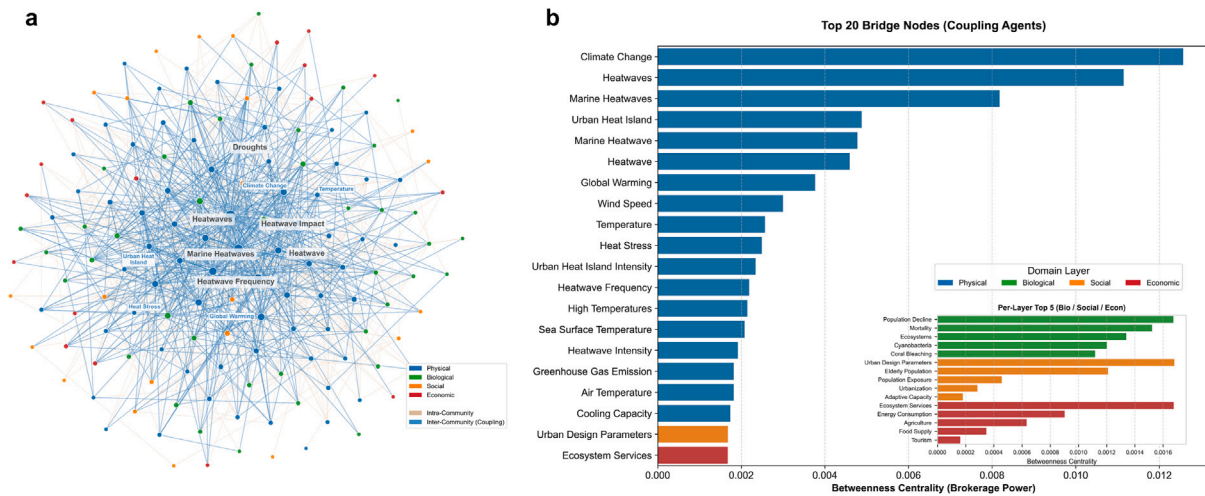


Fig. 4. Topological Anatomy of Cross-Sector Functional Coupling. (a) The network visualization displays 148 representative nodes (layer-balanced selection from 34,933) and 1021 edges, colored by domain layer (Physical, blue; Biological, green; Social, orange; Economic, red). Louvain community detection identifies 5 communities, with 536 inter-community edges (52.5%) indicating strong cross-sector coupling. (b) The main panel ranks the top-20 bridge nodes by betweenness centrality, revealing that 18 of 20 belong to the Physical layer, led by Climate Change (BC = 0.0126), Heatwaves (0.0111), and Marine Heatwaves (0.0082). The inset shows per-layer top-5 for non-Physical domains: Biological (Mortality, BC = 0.0033; Population Decline, 0.0033), Social (Elderly Population, 0.0017), and Economic (Energy Consumption, 0.0029). The absence of Biological entities from the global top-20 is consistent with their role as distributed receptors rather than centralized hubs.

reside not within individual sectors but at the latent interfaces where compound environmental extremes force a synchronization between natural ecosystems and human infrastructure.

4.4. Identification of latent transmission vectors

The topological analysis of the constructed knowledge graph facilitates the identification of high-impact low-frequency transmission vectors that typically evade detection in standard sectoral assessments. These latent cascading pathways represent long-tail risks characterized by high structural plausibility within the Earth system network despite occupying a marginal position in the existing literature, as illustrated in the scatter distribution of Fig. 5. Such discovered pathways suggest that systemic vulnerability is not uniformly distributed but disproportionately concentrated within specific bio-ecological mediation channels acting as invisible bridges between physical meteorological anomalies and socioeconomic instability. Examination of the extracted risk chains elucidates the mechanics of this cross-sectoral propagation where biological systems function as structural mediators of thermal stress. In the oceanic domain, the computational synthesis identifies a distinct cascading sequence whereby marine heatwaves act as drivers of disturbance, causing the mass mortality of foundation species such as kelp and seagrass. This computationally derived pathway aligns with global observations including the 2014–2016 Northeast Pacific warm anomaly and events in Western Australia, where the thermal exceedance of physiological thresholds caused the collapse of habitat-forming organisms (Wernberg et al., 2025; Smith et al., 2024; Holbrook et al., 2020). Such ecological degradation reshapes food web structures and erodes essential ecosystem services, thereby triggering substantial socioeconomic losses in fisheries and tourism sectors as evidenced by the closure of commercial scallop and crab fisheries following extreme thermal events (Free et al., 2023; Caputi et al., 2019). These findings suggest that the thermodynamic energy of oceanic warming propagates through complex intermediate shifts in marine biodiversity rather than translating directly into economic deficits. The analysis also reveals physiological feedback loops in terrestrial systems where the impact of extreme temperature is mediated by microscopic biological communities. Recent evidence corroborates the topological finding that heat

stress disrupts soil microbiome assembly and suppresses nitrogen-fixing bacteria, thereby compromising crop resilience and yield independent of water availability (Bei et al., 2023; Elango et al., 2025; Muhammad et al., 2024). This micro-scale vulnerability extends to human physiological health, where climate-induced alterations in gut microbiota composition exacerbate susceptibility to enteric pathogens and metabolic disorders (Litchman, 2025). To control for multiple comparisons, we applied a permutation test with 10,000 random-walk null paths and Benjamini–Hochberg FDR correction. The novelty scores of the discovered pathways are significantly higher than the null distribution (Mann–Whitney U , $p = 2.20 \times 10^{-105}$, Cohen's $d = 0.319$), and 526 of the 10,000 highest-scoring paths survive FDR correction at $\alpha = 0.05$, all originating from the high-novelty subset (novelty > 0.7; 526 of 4115 paths, 12.8%). The top-20 pathways each have empirical $p \leq 0.0003$; for instance, the top-ranked pathway exceeds 99.5% of null-distribution simulations, and the top-10 pathways collectively exceed 89.8%. These findings suggest that consequential climate risks reside in the blind spots of disciplinary silos, where the interplay between physical climatology and biological adaptation creates unexpected failure modes. Current adaptation strategies that focus solely on hardening physical infrastructure may underestimate the fragility of the living systems underpinning economic continuity.

4.5. Methodological robustness and evidence traceability

The bio-ecological mediation effect, cross-sector synchronization, and grey rhino pathways identified in Sections 4.2 and 4.4 challenge conventional linear risk assessment frameworks. To ensure that these macro-level findings are not artifacts of the knowledge extraction protocol, ontological design, or algorithmic parameters, we conducted a comprehensive robustness evaluation and edge-level provenance audit.

4.5.1. Ontology and data-source sensitivity

Table 2 compares six alternative layer schemes or assignment methods. All schemes for which BMR is defined yield $BMR > 1$, with even the most aggressive Social–Economic merger still producing $BMR = 4.419$. An alternative keyword-based layer-assignment rule (Cohen's $\kappa = 0.543$) yields $BMR = 7.645$, confirming that the mediation

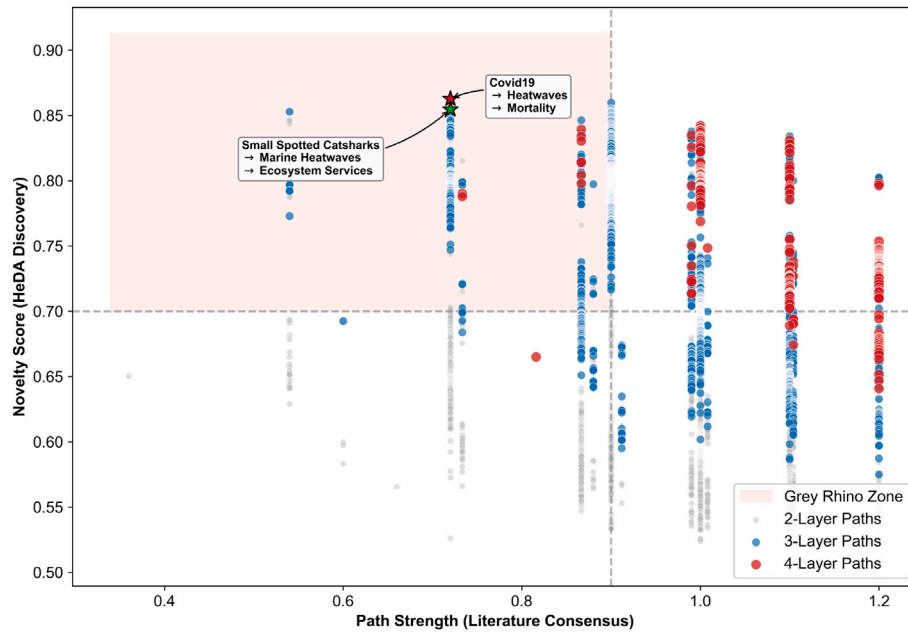


Fig. 5. Identification of Grey Rhino Risks in the Long Tail Distribution. The scatter plot contrasts path strength (literature consensus, x-axis) against novelty score (HeDA discovery, y-axis) for 10,000 cross-layer paths, colored by layer count (grey: 2-layer, $n = 6113$; blue: 3-layer, $n = 3533$; red: 4-layer, $n = 354$). The shaded Grey Rhino Zone (novelty > 0.7 , strength < 40 th percentile) contains 553 high-novelty but low-consensus paths (5.5% of total). Two representative pathways are annotated: (1) Covid-19 $\rightarrow$ Heatwaves $\rightarrow$ Mortality (novelty = 0.863, red star), illustrating compound biological-physical interactions; and (2) Small Spotted Catsharks $\rightarrow$ Marine Heatwaves $\rightarrow$ Ecosystem Services (novelty = 0.855, green star), representing an under-studied marine trophic cascade. Of all paths, 4115 (41.1%) exceed the novelty threshold of 0.7.

Table 2

Ontology sensitivity of the bio-ecological mediation ratio. Six alternative layer taxonomies or assignment methods are compared. BMR remains substantially above one under all schemes that preserve a biological mediating layer.

Layer scheme	Layers	BMR	Cross-layer %
Baseline (P/B/S/E)	4	8.231	52.0
Keyword-based rule ($\kappa = 0.543$)	4	7.645	51.8
Merge S + E $\rightarrow$ socioeconomic	3	4.419	51.0
Merge B + S $\rightarrow$ bio-social	3	9.278	48.2
Split B $\rightarrow$ health/eco/bio	6	8.231	55.4
Coarse-grain (env/human)	2	–	15.7

effect is not an artifact of the LLM classifier. The coarse two-layer partition (environment/human) collapses the mediating layer entirely, making BMR undefined by construction; however, its sharply reduced cross-layer proportion (15.7% vs. 52.0%) independently corroborates the structural importance of intermediate layers.

Because domain-specific publication volumes are highly imbalanced (Physical: 3892 papers; Biological: 1206; Social: 288; Economic: 76, a 51:1 ratio), raw edge counts may overrepresent well-studied domains. Table 3 re-computes BMR under three normalization schemes plus bootstrap resampling. Transition-probability and entity-count normalization both confirm $\text{BMR} > 1$, and the bootstrap 95% CI lower bound (7.457) remains far above unity. The only scheme under which BMR falls below one is paper-pair normalization, which inflates the denominator for the Economic layer by 51 $\times$ due to its minimal paper count; this artifact reflects a limitation of the normalizer rather than a reversal of the mediation pattern.

A stratified subsample of 80 papers (20 per layer) was processed under both abstract-only and full-text extraction (Table 4). Full-text extraction yields an order-of-magnitude increase in both total and cross-layer relations per paper, and shifts the layer distribution toward Physical entities ($\chi^2 = 60.7$, $p < 10^{-12}$). Crucially, the bio-ecological mediation ratio remains above one in both conditions, confirming that

Table 3

Literature-volume normalization of the Bio-ecological Mediation Ratio. Three normalization schemes and bootstrap resampling are compared. $\text{BMR} > 1$ holds under all methods except paper-pair normalization, whose failure is attributable to extreme layer-size imbalance.

Normalization scheme	BMR	$\text{BMR} > 1?$
Raw (no normalization)	8.231	Yes
Transition probability $P(\text{col} \text{row})$	8.999	Yes
Entity-count normalization	1.322	Yes
Paper-pair normalization	0.961	No ^a
Bootstrap mean (95% CI)	8.152 [7.457, 8.915]	Yes

^a E-layer has only 76 papers, inflating P $\rightarrow$ E density by 51 $\times$.

Bootstrap: 1000 iterations, document-level resampling with replacement.

Table 4

Abstract versus full-text knowledge-graph extraction ($N = 80$ stratified papers, 20 per layer).

Metric	Abstract	Full-text	Test
Relations/paper	7.3	86.5	Wilcoxon $p < 10^{-14}$
Cross-layer edges/paper	4.7	52.7	Wilcoxon $p < 10^{-14}$
Physical layer %	34.6	43.3	$\chi^2 = 60.7$, $p < 10^{-12}$
Biological layer %	29.5	29.2	n.s.
BMR	2.91	2.60	Both > 1

abstract-based extraction does not introduce a systematic bias in the direction of the mediation finding.

4.5.2. Algorithmic parameter stability

We swept 55 combinations of (α, β, γ) weights ($\alpha + \beta + \gamma = 1$, $\alpha \in [0.3, 0.7]$, step 0.05). Pairwise Spearman ρ across all combinations averages 0.876 (all $p < 5.57 \times 10^{-97}$), and Kendall's coefficient of concordance $W = 0.806$ ($\chi^2 = 443,445$, $p < 5 \times 10^{-324}$). The mean Jaccard overlap of the Top-20 pathway set is 0.517, with 20 pathways appearing in the Top-50 under more than 80% of weight combinations, indicating that the grey rhino ranking is robust to weight perturbation.

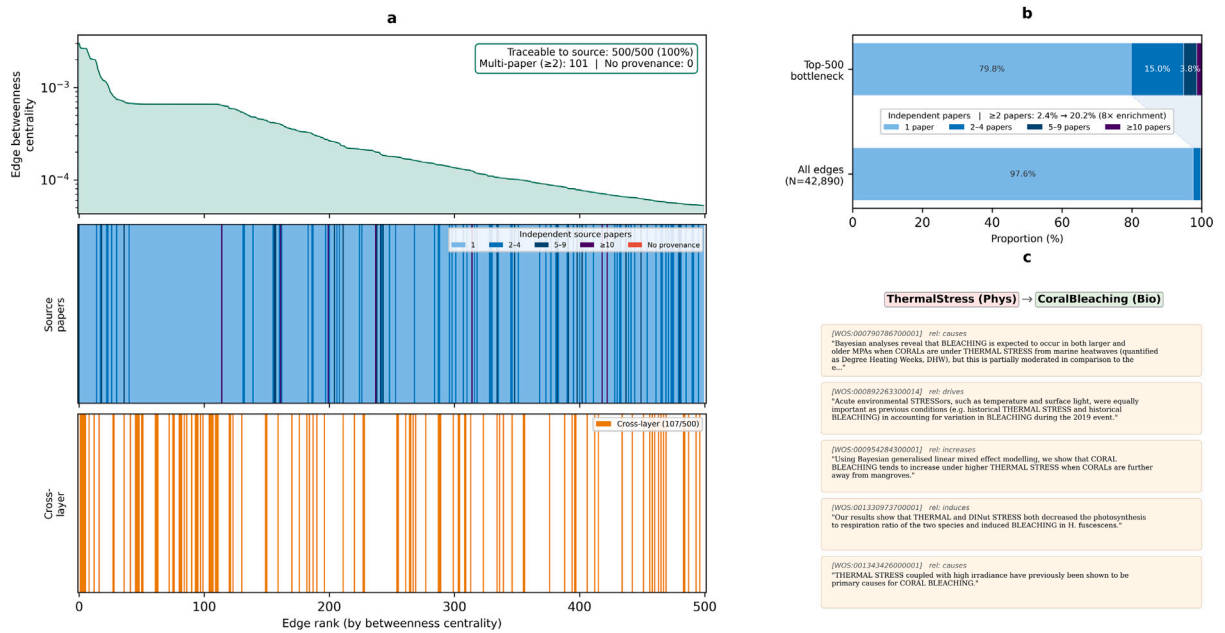


Fig. 6. Edge-Level Provenance Validation. (a) Provenance profile of the top-500 bottleneck edges ranked by edge betweenness centrality (BC). The upper curve shows BC decay; the middle barcode is colored by the number of independent source papers (1 light blue, 2–4 blue, 5–9 dark blue, ≥ 10 purple); the bottom strip marks cross-layer edges (orange, 101/500). All 500 bottleneck edges are traceable to source papers (100%), with 101 (20.2%) corroborated by ≥ 2 independent publications. (b) Consensus enrichment of multi-paper corroboration. 100% stacked bars compare the paper-count distribution between all 42,890 edges and the top-500 bottleneck edges. The proportion supported by ≥ 2 papers increases from 2.4% globally to 20.2% among bottleneck edges (8x enrichment). (c) Evidence penetration example: ThermalStress (Phys) $\rightarrow$ CoralBleaching (Bio). This cross-layer edge is supported by 5 independent papers, demonstrating multi-source validation of the same relationship from different perspectives (causes, drives, increases, induces).

Table 5

Sensitivity of topological metrics to entity deduplication thresholds. Production threshold ($d < 0.03$, bold) merges only 2.2% of entities while preserving rank stability (Kendall $\tau = 0.876$). BMR remains near 8.1 across all configurations.

	No-merge	$d < 0.02$	$d < 0.03$	$d < 0.05$	$d < 0.08$	$d < 0.10$
Merge rate (%)	0	0.9	2.2	5.9	13.3	19.7
Precision	–	0.701	0.699	0.567	0.290	0.237
Kendall τ	1.000	0.968	0.876	0.366	–0.082	–0.129
BMR	8.148	8.144	8.142	8.139	8.154	8.127
Cross-layer %	0.525	0.525	0.525	0.525	0.525	0.526

Synonym merging via embedding-based clustering is a potential source of topological distortion through over-merging (false synonymy creating artificial hubs) or under-merging (fragmentation inflating node counts). We evaluated five cosine-distance thresholds ($d < 0.02, 0.03, 0.05, 0.08, 0.10$) against a no-merge baseline using LLM-judged precision (300 merged pairs per threshold) and Kendall τ rank correlation of betweenness centrality (Table 5). The production threshold $d < 0.03$ merges 2.2% of entities with a precision of 0.699 and Kendall $\tau = 0.876$, reflecting a conservative strategy that favors under-merging over false synonymy. Crucially, the bio-ecological mediation ratio is virtually invariant across all six configurations (range 8.127–8.154), and the cross-layer edge proportion remains at 0.525–0.526, demonstrating that the mediation finding is insensitive to the deduplication parameter.

4.5.3. Edge-level traceability

Every edge in the knowledge graph is stored with its source paper identifier and verbatim evidence sentence, yielding 100% traceability for all 42,890 edges and full edge-level coverage for all 4115 Grey Rhino pathways. To assess whether structurally important edges are supported by convergent evidence, we examined the top-500 edges

ranked by betweenness centrality. Among these bottleneck edges, 20.2% are corroborated by two or more independent publications, compared with 2.4% globally—an eight-fold enrichment (Fig. 6a–b). These 500 edges draw on 1728 independent source papers. Fig. 6c illustrates the provenance profile for a representative cross-layer edge (ThermalStress $\rightarrow$ CoralBleaching, Physical $\rightarrow$ Biological), which is independently supported by five papers using distinct relational descriptors (causes, drives, increases, induces). A full audit of the top-20 Grey Rhino pathways with edge-level citations is provided in Table 6.

5. Discussion & conclusion

The escalating frequency of compound heatwaves calls for a reassessment of risk assessment frameworks, as the non-linear cascading effects of thermal stress increasingly transcend the resolution boundaries of traditional coupled physical-biogeochemical models (Kornhuber et al., 2024; Sutanto et al., 2025; Schillerberg and Tian, 2024). While individual meteorological hazards are well characterized, the propagation of these anomalies through interconnected human and natural systems generates systemic failures that are more severe than the sum of discrete stressors (Pescaroli et al., 2024; Bevacqua et al., 2023; Duvat, 2026). Our topological reconstruction of the global risk landscape suggests that the interactions between atmospheric dynamics and socioeconomic stability are governed by complex feedback loops that defy linear impact assessments. By synthesizing fragmented mechanisms into a unified risk topology, this study suggests that the key vulnerabilities lie not within individual sectors but at the interfaces where physical climate extremes force a functional synchronization between natural ecosystems and critical infrastructure. HeDA's architecture is model-agnostic. While we instantiate the default pipeline with Qwen3-Max for cost-efficiency (achieving an overall accuracy

Table 6

Edge-level provenance audit for the top-20 Grey Rhino pathways (ranked by novelty score). Each propagation step is traced to its source paper and a representative evidence sentence. N = number of independent papers supporting the edge.

Step	Propagation edge	Rel.	Domain shift	N	Representative evidence (excerpt)	WOS accession
#1 (Score = 0.863, LF = 0.969): Covid19 → Heatwaves → Mortality						
1	Covid19 → Heatwaves	compoundsImpactWith	Bio → Phys	1	“COVID-19 and heatwaves: a double whammy for Indian cities”	WOS:000573750600007
2	Heatwaves → Mortality	associatedWith	Phys → Bio	30	“Recognizing the importance of specific climatic events such as heatwaves, systematic documentation and asse...”	WOS:000383615200014
#2 (Score = 0.862, LF = 0.968): CommunityBasedHealthProfessionOrganisations → Heatwaves → Mortality						
1	CommunityBasedHealth-ProfessionOrganisations → Heatwaves	minimisesHarmFrom	Soc → Phys	1	“MINIMISING HARM TO ELDERLY VICTORIANS FROM HEATWAVES: A QUALITATIVE STUDY OF THE ROLE OF COMMUNITY-BASED HE...”	WOS:000284077200305
2	Heatwaves → Mortality	associatedWith	Phys → Bio	30	“Recognizing the importance of specific climatic events such as heatwaves, systematic documentation and asse...”	WOS:000383615200014
#3 (Score = 0.860, LF = 0.957): AdaptiveCapacity → Heatwaves → Mortality						
1	AdaptiveCapacity → Heatwaves	allowsSurvivalOf	Soc → Phys	1	“The integration of molecular, physiological, and behavioral responses within a species illustrates the kind...”	WOS:000645224200016
2	Heatwaves → Mortality	associatedWith	Phys → Bio	30	“Recognizing the importance of specific climatic events such as heatwaves, systematic documentation and asse...”	WOS:000383615200014
#4 (Score = 0.855, LF = 0.984): AlzheimersDisease → Heatwaves → ElderlyPopulation						
1	AlzheimersDisease → Heatwaves	posesHigherRiskDuring	Bio → Phys	1	“Alzheimer’s disease (AD) and other dementia patients face a higher risk of mortality during heatwaves, but ...”	WOS:001332119000001
2	Heatwaves → ElderlyPopulation	affectsVulnerability	Phys → Soc	6	“Older people and those living with frailty are more vulnerable to the effects of climate change including h...”	WOS:000776252000055
#5 (Score = 0.855, LF = 0.983): IncreasedEnergyPrices → ElderlyPopulation → Heatwaves						
1	IncreasedEnergyPrices → ElderlyPopulation	increasesVulnerabilityOf	Econ → Soc	1	“There will be sections of the community, such as the elderly and the less well off, that will be particular...”	WOS:000313387800004
2	ElderlyPopulation → Heatwaves	hasVulnerabilityTo	Soc → Phys	2	“Particularly vulnerable to heatwaves are elderly people with limited mobility, who have limited means to se...”	WOS:000595067700017
#6 (Score = 0.855, LF = 0.983): Covid19 → Heatwaves → ElderlyPopulation						
1	Covid19 → Heatwaves	compoundsImpactWith	Bio → Phys	1	“COVID-19 and heatwaves: a double whammy for Indian cities”	WOS:000573750600007
2	Heatwaves → ElderlyPopulation	affectsVulnerability	Phys → Soc	6	“Older people and those living with frailty are more vulnerable to the effects of climate change including h...”	WOS:000776252000055
#7 (Score = 0.855, LF = 0.977): AdaptiveCapacity → Heatwaves → ElderlyPopulation						
1	AdaptiveCapacity → Heatwaves	allowsSurvivalOf	Soc → Phys	1	“The integration of molecular, physiological, and behavioral responses within a species illustrates the kind...”	WOS:000645224200016
2	Heatwaves → ElderlyPopulation	affectsVulnerability	Phys → Soc	6	“Older people and those living with frailty are more vulnerable to the effects of climate change including h...”	WOS:000776252000055
#8 (Score = 0.855, LF = 0.983): CommunityBasedHealthProfessionOrganizations → Heatwaves → ElderlyPopulation						
1	CommunityBasedHealth-ProfessionOrganizations → Heatwaves	minimizesHarmFrom	Soc → Phys	1	“MINIMISING HARM TO ELDERLY VICTORIANS FROM HEATWAVES: A QUALITATIVE STUDY OF THE ROLE OF COMMUNITY-BASED HE...”	WOS:000284077200305
2	Heatwaves → ElderlyPopulation	affectsVulnerability	Phys → Soc	6	“Older people and those living with frailty are more vulnerable to the effects of climate change including h...”	WOS:000776252000055
#9 (Score = 0.855, LF = 0.991): AdaptiveCapacity → Heatwaves → Stakeholders						
1	AdaptiveCapacity → Heatwaves	allowsSurvivalOf	Soc → Phys	1	“The integration of molecular, physiological, and behavioral responses within a species illustrates the kind...”	WOS:000645224200016
2	Heatwaves → Stakeholders	affectPerceptionOf	Phys → Soc	1	“Perception and Response to Heatwaves in Adelaide: A Qualitative Interview About the Elderly and Stakeholders”	WOS:000285400800020

(continued on next page)

Table 6 (continued).

Step	Propagation edge	Rel.	Domain shift	N	Representative evidence (excerpt)	WOS accession
#10 (Score = 0.855, LF = 0.986): SmallSpottedCatsharks → MarineHeatwaves → EcosystemServices						
1	SmallSpottedCatsharks → MarineHeatwaves	copeWith	Bio → Phys	1	“Small-spotted catsharks cope with marine heatwaves, for now”	WOS:001408640500009
2	MarineHeatwaves → EcosystemServices	disrupt	Phys → Econ	5	“MHWs, which will probably intensify with anthropogenic climate change(3), are rapidly emerging as forceful ...”	WOS:000462353800016
#11 (Score = 0.855, LF = 0.991): AdaptiveCapacity → Heatwaves → ElderlyVictorians						
1	AdaptiveCapacity → Heatwaves	allowsSurvivalOf	Soc → Phys	1	“The integration of molecular, physiological, and behavioral responses within a species illustrates the kind...”	WOS:000645224200016
2	Heatwaves → ElderlyVictorians	causesHarmTo	Phys → Bio	1	“MINIMISING HARM TO ELDERLY VICTORIANS FROM HEATWAVES: A QUALITATIVE STUDY OF THE ROLE OF COMMUNITY-BASED HE...”	WOS:000284077200305
#12 (Score = 0.854, LF = 0.988): AdaptiveCapacity → Heatwaves → OlderPeople						
1	AdaptiveCapacity → Heatwaves	allowsSurvivalOf	Soc → Phys	1	“The integration of molecular, physiological, and behavioral responses within a species illustrates the kind...”	WOS:000645224200016
2	Heatwaves → OlderPeople	causeHarmTo	Phys → Soc	1	“To investigate the role of community-based health profession and carer organizations in Victoria in minimis...”	WOS:000281522800004
#13 (Score = 0.854, LF = 0.989): AdaptiveCapacity → Heatwaves → OlderPersons						
1	AdaptiveCapacity → Heatwaves	allowsSurvivalOf	Soc → Phys	1	“The integration of molecular, physiological, and behavioral responses within a species illustrates the kind...”	WOS:000645224200016
2	Heatwaves → OlderPersons	causesHarmTo	Phys → Soc	3	“MINIMISING HARM TO OLDER PERSONS FROM HEATWAVES: A SURVEY OF THE AWARENESS, KNOWLEDGE AND PRACTICES OF COMM...”	WOS:000284077200304
#14 (Score = 0.854, LF = 0.990): AdaptiveCapacity → Heatwaves → ElderlyPeople						
1	AdaptiveCapacity → Heatwaves	allowsSurvivalOf	Soc → Phys	1	“The integration of molecular, physiological, and behavioral responses within a species illustrates the kind...”	WOS:000645224200016
2	Heatwaves → ElderlyPeople	posesHealthThreatTo	Phys → Soc	1	“The increasing risk of heatwaves in England poses a particular threat to the health of elderly people.”	WOS:000272184900007
#15 (Score = 0.854, LF = 0.982): PriorSaltExposure → Heatwaves → ElderlyPopulation						
1	PriorSaltExposure → Heatwaves	causesCommunitiesTo-RespondDifferentlyTo	Bio → Phys	1	“As a result, communities recovering from previous salt exposure responded differently to heatwaves than und...”	WOS:001236082500001
2	Heatwaves → ElderlyPopulation	affectsVulnerability	Phys → Soc	6	“Older people and those living with frailty are more vulnerable to the effects of climate change including h...”	WOS:000776252000055
#16 (Score = 0.854, LF = 0.985): NativeCanopyFormingKelps → MarineHeatwaves → EcosystemServices						
1	NativeCanopyFormingKelps → MarineHeatwaves	declinedAfter	Bio → Phys	1	“Turf-forming seaweeds, especially invasive seaweeds, generally increased in abundance after a MHW, whereas ...”	WOS:000502973700001
2	MarineHeatwaves → EcosystemServices	disrupt	Phys → Econ	5	“MHWs, which will probably intensify with anthropogenic climate change(3), are rapidly emerging as forceful ...”	WOS:000462353800016
#17 (Score = 0.854, LF = 0.985): NativeCanopyFormingFucoids → MarineHeatwaves → EcosystemServices						
1	NativeCanopyForming-Fucoids → MarineHeatwaves	declinedAfter	Bio → Phys	1	“Turf-forming seaweeds, especially invasive seaweeds, generally increased in abundance after a MHW, whereas ...”	WOS:000502973700001
2	MarineHeatwaves → EcosystemServices	disrupt	Phys → Econ	5	“MHWs, which will probably intensify with anthropogenic climate change(3), are rapidly emerging as forceful ...”	WOS:000462353800016
#18 (Score = 0.854, LF = 0.989): AdaptiveCapacity → Heatwaves → BuildingResilienceConcerns						
1	AdaptiveCapacity → Heatwaves	allowsSurvivalOf	Soc → Phys	1	“The integration of molecular, physiological, and behavioral responses within a species illustrates the kind...”	WOS:000645224200016
2	Heatwaves → BuildingResilienceConcerns	inform	Phys → Soc	1	“There is concern about the resilience of these buildings in a changing climate, informed by the experience ...”	WOS:000300758600007

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Table 6 (continued).

Step	Propagation edge	Rel.	Domain shift	N	Representative evidence (excerpt)	WOS accession
#19 (Score = 0.854, LF = 0.980): MaleReproductiveBiology → Heatwaves → ElderlyPopulation						
1	MaleReproductiveBiology → Heatwaves	isVulnerableTo	Bio → Phys	1	"Recent data from several species, including the beetle Tribolium castaneum, confirmed that male reproductiv..."	WOS:001329779400001
2	Heatwaves → ElderlyPopulation	affectsVulnerability	Phys → Soc	6	"Older people and those living with frailty are more vulnerable to the effects of climate change including h..."	WOS:000776252000055
#20 (Score = 0.853, LF = 0.950): ThermallyInefficientDwellings → Heatwaves → Mortality						
1	ThermallyInefficient-Dwellings → Heatwaves	increasesVulnerabilityTo	Econ → Phys	1	"Energy-poor households in Africa's burgeoning urban informal settlements are especially likely to suffer fr..."	WOS:000601608500003
2	Heatwaves → Mortality	associatedWith	Phys → Bio	30	"Recognizing the importance of specific climatic events such as heatwaves, systematic documentation and asse..."	WOS:000383615200014

of 61.18%), substituting other foundation models such as Claude-Sonnet-4.5 or GPT-5.4 yields consistently robust performance (59.89% and 58.79%, respectively). This consistent improvement suggests that the knowledge-augmentation paradigm generalizes across foundation models rather than depending on a specific provider.

A central finding of this analysis is the identification of a dominant bio-ecological mediation effect, which contributes to understanding how thermodynamic energy translates into systemic loss. Topological analysis suggests that biological systems, specifically agricultural productivity and human physiological health, function as the primary structural mediators of environmental hazards. The structural topology indicates that, in the modeled risk landscape, physical atmospheric anomalies are rarely represented as propagating directly to economic deficits but are instead mediated through the degradation of these biological assets, which act as a saturation bottleneck in the global risk network. This topological feature provides a structural explanation for why compound events such as heatwaves coinciding with droughts generate disproportionate impacts on food security and labor capacity (Tabari and Willems, 2023; Hassan et al., 2026) effectively bridging the theoretical gap between meteorological observation and macroeconomic instability. Consequently, adaptation strategies that focus exclusively on hardening physical infrastructure systematically underestimate systemic vulnerability by neglecting the fragility of the living systems that underpin economic continuity, a gap that nature-based solutions are increasingly proposed to address (Seddon, 2022; Turner et al., 2022; Vlachogiannis et al., 2026).

The topological analysis further elucidates the emergence of synchronized failure modes across theoretically distinct sectors where independent systems coalesce into tightly entangled risk communities under extreme thermal regimes. We observe that compound hazards such as marine heatwaves and hydrological droughts act as coupling agents that dissolve the functional boundaries between energy distribution networks and emergency medical services. This structural interdependence implies that the thermodynamic failure of cooling mechanisms can rapidly cascade to clinical capacity saturation (Stone Jr. et al., 2023; Korfmacher et al., 2025), creating a cross-domain vulnerability that is unlikely to be adequately addressed through isolated sectoral planning. These findings identify grey rhino risks located in the long tail of the probability distribution, which represent structurally plausible but underexplored transmission vectors connecting oceanic and atmospheric processes to terrestrial socioeconomic outcomes. However, the current framework is structurally rather than temporally oriented; while the knowledge graph captures directional associations reported in the literature, it does not model the timing, sequencing, or lag structure of these cascading failures. Integrating event-level temporal data such as EM-DAT disaster timelines and reanalysis-driven exposure indices (Hersbach et al., 2020) with the topological skeleton would enable future work to advance from structural hypothesis

generation toward temporally resolved verification. Similarly, the directed edges reflect associative co-occurrence as described by source authors rather than statistically identified causal effects; formal causal inference through structural equation modeling or mediation analysis with observational climate data remains a necessary next step. Finally, all relational triplets are extracted from English-language abstracts indexed in the Web of Science, and non-English literature or evidence embedded in full-text supplementary materials may be underrepresented, although our abstract-versus-full-text sensitivity test (Section 4.5.1) indicates that the directional mediation finding is robust to this limitation.

In conclusion, this study advances the research on compound weather and climate extremes by establishing a data-driven framework for deciphering the anatomy of systemic risk propagation. Using AI as an integrative tool, we synthesize disparate scientific evidence to uncover the bio-ecological bottlenecks and latent functional couplings that characterize the Anthropocene risk landscape. These insights suggest that effective climate adaptation must evolve from static defense against discrete weather events to a dynamic resilience approach (Zorita et al., 2025) that explicitly strengthens the biological mediators and manages the connectivity between coupled human-natural systems. As the interactions between land-atmosphere processes and societal feedback loops intensify, the ability to anticipate these non-linear cascading pathways will be important for improving resilience as thermal extremes intensify.

CRediT authorship contribution statement

Yiquan Wang: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tin-Yeh Huang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Qingyun Gao:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Yuhan Chang:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation. **Jialin Zhang:** Writing – review & editing, Writing – original draft, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source corpus was retrieved from the Web of Science Core Collection using the topic query “heatwave” (covering title, abstract, and keywords; all document types and languages included; retrieval date: 8 April 2026), yielding 8,365 records spanning 1928–2026. Dataset S1 (Dataset_S1_WOS_Records.csv) provides the full list of Web of Science accession numbers, titles, publication years, journal names, DOIs, and temporal-split assignments (KG construction set, ≤ 2024 , $n = 5,739$; held-out evaluation set, ≥ 2025 , $n = 2,626$). All analysis code, the knowledge-graph construction pipeline, and supplementary datasets are available at <https://github.com/wyqmath/heatwave>.

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